

→ The fuse wire is connected to the _____ wire.

→ If I had to choose a fuse wire for a device with a greater power, I would select a wire with greater thickness or lesser thickness.

$$P = VI$$

$$\underline{I \propto P}$$

→ A device is rated 220V - 1500W. Find the safe current limit for its fuse

$$P = VI$$

$$1500 = 220 \times I$$

$$I = \frac{1500}{220} = \frac{75}{11} = 6.81 \text{ A}$$

∴ Safe current must be $> 6.81 \text{ A}$ (8 A, 9 A, 10 A) 1

Two fuse wires are rated 5 A and 20 A. Which of the two is (i) thicker, (ii) longer?

→ 20 A → thicker. (thicker — less resistance — more current)
5 A → longer (longer — more resistance — less current)

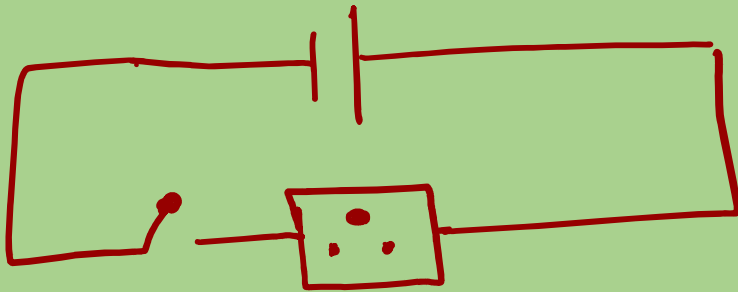
'A fuse is rated 8 A'. Can it be used with an electrical appliance of rating 5 kW, 200 V?

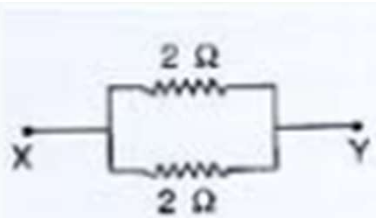
→ $P = VI$ → draws 25 A.
 $5 \times 1000 = 200 \times I$ ∴ A fuse of 8 A can not be used, as
 $\Rightarrow I = \frac{5000}{200}$ it will be blown off by the
 $= 25 \text{ A}$ high current.

An electric kettle is rated 3 kW, 250 V. Give reason whether this kettle can be used in a circuit which contains a fuse of current rating 13 A

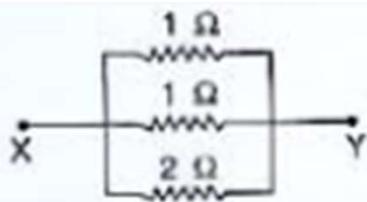
⇒ $P = VI$ | $3000 = 250 \times I$
 $\therefore I = 12 \text{ A} \therefore \text{Yes.}$

→ why switches are always connected to live?

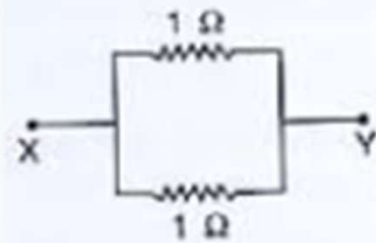




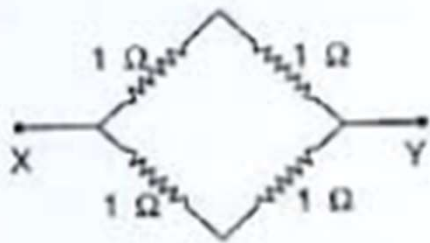
(a)



(b)



(c)



(d)

b

$$\frac{1}{R} = \frac{1}{1} + \frac{1}{1} + \frac{1}{2}$$

$$\frac{1}{R} = \frac{2+2+1}{2}$$

$$= \frac{5}{2}$$

$$\therefore R = \frac{2}{5} = 0.4 \Omega$$

a

$$\frac{1}{R} = \frac{1}{2} + \frac{1}{2}$$

$$= \frac{1+1}{2}$$

$$\therefore R = 1 \Omega$$

$\therefore$ a and d.

c

$$\frac{1}{R} = \frac{1}{1} + \frac{1}{1}$$

$$= \frac{1+1}{1}$$

$$\frac{1}{R} = 2$$

$$\therefore R = \frac{1}{2}$$

$$= 0.5 \Omega$$

d

$$R_s = 1 + 1 = 2 \Omega$$

$$R_s = 1 + 1 = 2 \Omega$$

$$\frac{1}{R} = \frac{1}{2} + \frac{1}{2}$$

$$\therefore \frac{1}{R} = \frac{2}{2}$$

$$\therefore R = 1 \Omega$$

The music system draws a current of 400mA when connected to a 12V battery.

- (i) What is the resistance of the music system
(ii) The music system is left playing for several hours and finally the battery voltage drops and the music system stops playing when the current drops to 320mA. At what battery voltage does the music system stop playing?

$$I = 400 \text{ mA} = \frac{400}{1000} = 0.4 \text{ A}$$

$$V = 12 \text{ V}$$

(i) $V = IR$

$$12 = 0.4 \times R$$

$$\therefore R = 30 \Omega$$

(ii) $I = 320 \text{ mA} = 0.32 \text{ A}$

$$V = IR$$

$$= 0.32 \times 30$$

$$= 9.6 \text{ V}$$

< 9.6V is not enough to run this system..

An electric geyser rated **1500 W** is used for **2 hours** daily. In the same house, **four** LED bulbs, each rated **9 W**, are used for **5 hours** daily.

Calculate the following:

1. The total electrical energy consumed in one day in commercial units (kWh).
2. The total electrical energy consumed in one day in SI units (Joules).
3. The total electricity bill for the month of April (~~2020~~) if the cost of electricity is **₹ 6.00 per unit**.

$$P_1 = 1500 \text{ W} \quad t_1 = 2 \text{ h}$$

$$W = Pt = VIt = \frac{V^2 t}{R}$$

$$P_2 = 9 \text{ W}, \quad t_2 = 5 \text{ h}$$

$$\text{Total energy} = (3 + 0.18) = 3.18 \text{ kWh.}$$

$$\cdot \text{Geyser} = (1500 \times 2) = 3000 \text{ W} = 3 \text{ kW}$$

$$\cdot \text{bulb} = (9 \times 5 \times 4) = 18 \text{ W} = 0.18 \text{ kW}$$

$$1 \text{ kWh} = 3.6 \times 10^6 \text{ J}$$

$$\therefore 3.18 \text{ kWh} = 3.6 \times 10^6 \times 3.18 \text{ J} = 11.45 \times 10^6 \text{ J.}$$

$$APR = 30 \text{ d.}$$

Consumed -

$$\begin{aligned}\therefore \text{Total energy, in one month} &= (3.18 \times 30) \\ &= 9.54 \times 10 \\ &= 95.4 \text{ kWh}\end{aligned}$$

$$\therefore \text{Cost} = (95.4 \times 6) = ₹572.4.$$